INNOVATION PROJECT PRESENTATION

FOOTSTEP POWER GENERATION USING ARDUINO

Converting Every Step into Sustainable Energy (1-Meter Walkway Project)

Presented By: Department of Computer Science

PROJECT IDENTIFICATION

PROJECT TEAM DETAILS

24BCA7086
BALANJANEYA MANIKANTA KILARI

24BCA7337
KONJETI PREM CHAND

24BCA7473
KARANAM MAHESH

24BCA7347
PADITAPU AVINASH

24BCA7294
MUNAGALA DOLA GURU SAI PAVAN

24BCA7572
BHAIRAPURAM MOHAMMED OWAIZ

ACADEMIC SUPERVISION

PROJECT GUIDE INFORMATION



PROJECT GUIDE

Kosuri Yellareswararao

EMPLOYEE ID

70656

DESIGNATION

Assistant Professor Sr. Grade-2

SCHOOL / DEPARTMENT

School of Advanced Sciences (SAS)

CIRCUIT SCHEMATICS & WIRING DIAGRAM

1-METER FOOTSTEP POWER PROJECT WIRING

24x Piezo Sensor Matrix (1m)

- 24× 35mm PZT Discs across 1m Walkway
- Series-Parallel Grid Network
- Maximum Mechanical Compression
- High Voltage Pulsed Output

Rectifier & Filter Network

- 16× 1N4007 Bridge Diodes Array
- 4–6× 10µF Smoothing Capacitors
- 10kΩ / 100kΩ Voltage Divider

Arduino Uno Microcontroller

- Analog Pin A0 (Voltage Sensing)
- Digital Pin 8 (Step Sensor Signal)
- Pins A4 (SDA) | A5 (SCL) - I2C Bus
- ATmega328P Core Logic Processor

Transistor Switching & Battery Charging Module

BC547 NPN Switch:

- Collector → Digital Pin 8
- Base → 10k Resistor Logic
- Emitter → GND Line

Li-ion Battery Protection:

- 2× 18650 Li-ion Cells (2-Cell Holder)
- Dedicated BMS Charging Protection
- VCC / Vin → Common System GND

16x2 I2C LCD Output

```
1M TILE STEP: 1
VOLTAGE: 11.94mV
```

16×2 I2C Display (GND, VCC, SDA, SCL)

TECHNICAL ARCHITECTURE PICTURE

HARDWARE & SENSOR LAYER STRUCTURE

TOP TREAD TILE (1m × 0.5m ACRYLIC & MDF WOOD BASE)

Converts Dynamic Human Foot Weight Across Walkway → Mechanical Compression

↓ ↓ ↓

Piezo Array 1 (8 Discs)

35mm PZT Matrix Grid

Piezo Array 2 (8 Discs)

35mm PZT Matrix Grid

Piezo Array 3 (8 Discs)

35mm PZT Matrix Grid

↓ ↓ ↓

Full-Bridge Rectifier

16× 1N4007 Diodes + Capacitors
(Converts Pulsed AC → DC)

→

Li-ion Battery Bank

2× 18650 Cells + Protection Module
(Stores Harvested Energy)

→

Arduino & 16x2 LCD

ATmega328P Microcontroller
(Voltage & Step Monitoring)

HARDWARE SPECIFICATIONS

1-METER PROJECT COMPONENT SHOPPING LIST

Mechanical Components

1. Acrylic top sheet (1 m × 0.5 m)	1 Pc
2. MDF / Wood base (1 m × 0.5 m)	1 Pc
3. Spacers / Springs	8–12 Pcs
4. Rubber pads	8–12 Pcs
5. Screws, nuts and bolts	1 Set
6. Glue gun + Glue sticks	1 Set
7. Adhesive / Heavy-duty tape	1 Roll

Electrical Components

8. 35 mm Piezo Discs	24 Pcs
9. 1N4007 Rectifier Diodes	16 Pcs
10. 10 µF Electrolytic Capacitors	4–6 Pcs
11. Resistors (10 kΩ, 100 kΩ & assorted)	1 Pack
12. BC547 NPN Transistors	1–2 Pcs
13. Arduino UNO Board	1 Pc
14. 16×2 LCD Display with I2C Module	1 Pc
15. Mini Breadboard & Jumper/Hookup Wires	1 Set
16. 18650 Li-ion Cells (2-Cell Holder + BMS)	1 Set

OVERVIEW

INTRODUCTION TO FOOTSTEP POWER GENERATION

Renewable Energy Need

The increasing global demand for electrical energy and rapid depletion of fossil fuels create an urgent necessity for sustainable energy sources.

Wasted Mechanical Energy

Every single day, thousands of footsteps generate substantial mechanical energy in public spaces that normally goes completely uncaptured.

Piezoelectric Technology

This project utilizes piezoelectric sensors integrated within a 1-meter walkway tile and Arduino microcontroller to harvest foot pressure into electrical power.

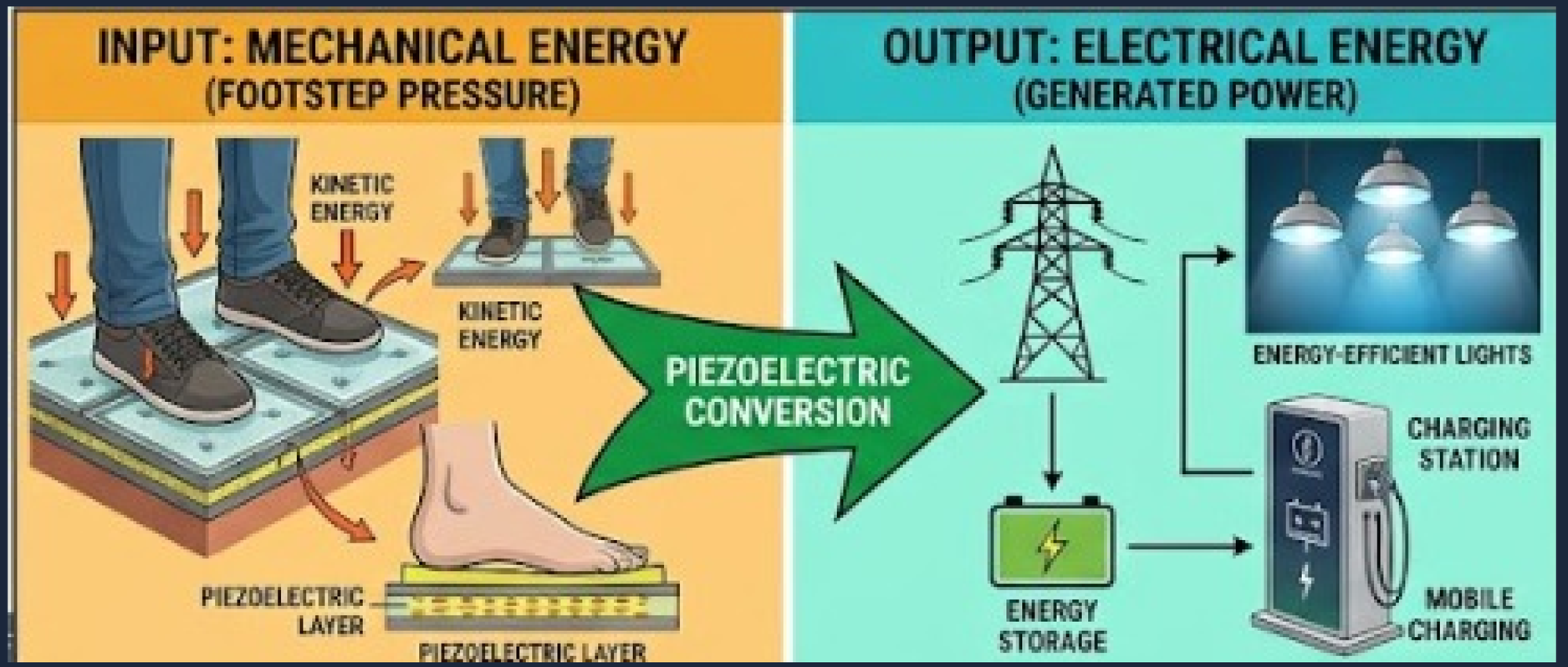
Energy Storage & Utility

Harvested energy is regulated and stored in rechargeable batteries to power low-energy IoT devices, emergency lights, and sensors.

FOOTSTEP POWER GENERATION

HOW FOOTSTEPS BECOME ELECTRICITY

MECHANICAL ENERGY → PIEZOELECTRIC CONVERSION → ELECTRICAL OUTPUT



The diagram directly supports the project flow shown on the existing process-flow slide.

Additional project visual

CHALLENGE & CONTEXT

PROBLEM STATEMENT

01 Electricity demand is increasing rapidly worldwide.

02 Conventional energy resources are limited and ecologically harmful.

03 Human walking generates mechanical energy that is normally wasted.

04 Public places (railway stations, malls, airports, colleges) experience thousands of footsteps every day.

05 There is a compelling need to utilize this wasted mechanical energy effectively.

PROCESS FLOW DIAGRAM

STEP-TO-POWER ENERGY CONVERSION FLOW

1. Mechanical Pressure

- Human foot pressure on 1m tile
- Force: 50kg - 80kg per step
- High footfall public zones

2. Piezo Transduction

- 24x PZT crystals convert strain
- Generates variable AC voltage
- Parallel/series sensor grid

3. Signal Conditioning

- Bridge rectifier & capacitor
- Smooths AC signal into DC
- Regulated charge output

4. Storage & Monitoring

- Rechargeable 18650 battery
- Arduino Uno voltage sensing
- Real-time 16×2 LCD status

PROJECT GOALS

OBJECTIVES

- Generate electrical power from human footsteps.
- Convert mechanical pressure into electrical energy using piezoelectric sensors.
- Monitor generated voltage and step counts accurately using Arduino.
- Store the generated energy safely in rechargeable Li-ion batteries.
- Display real-time voltage and step metrics on an I2C 16×2 LCD screen.
- Promote renewable and sustainable energy solutions for smart cities.

COMPARATIVE ANALYSIS

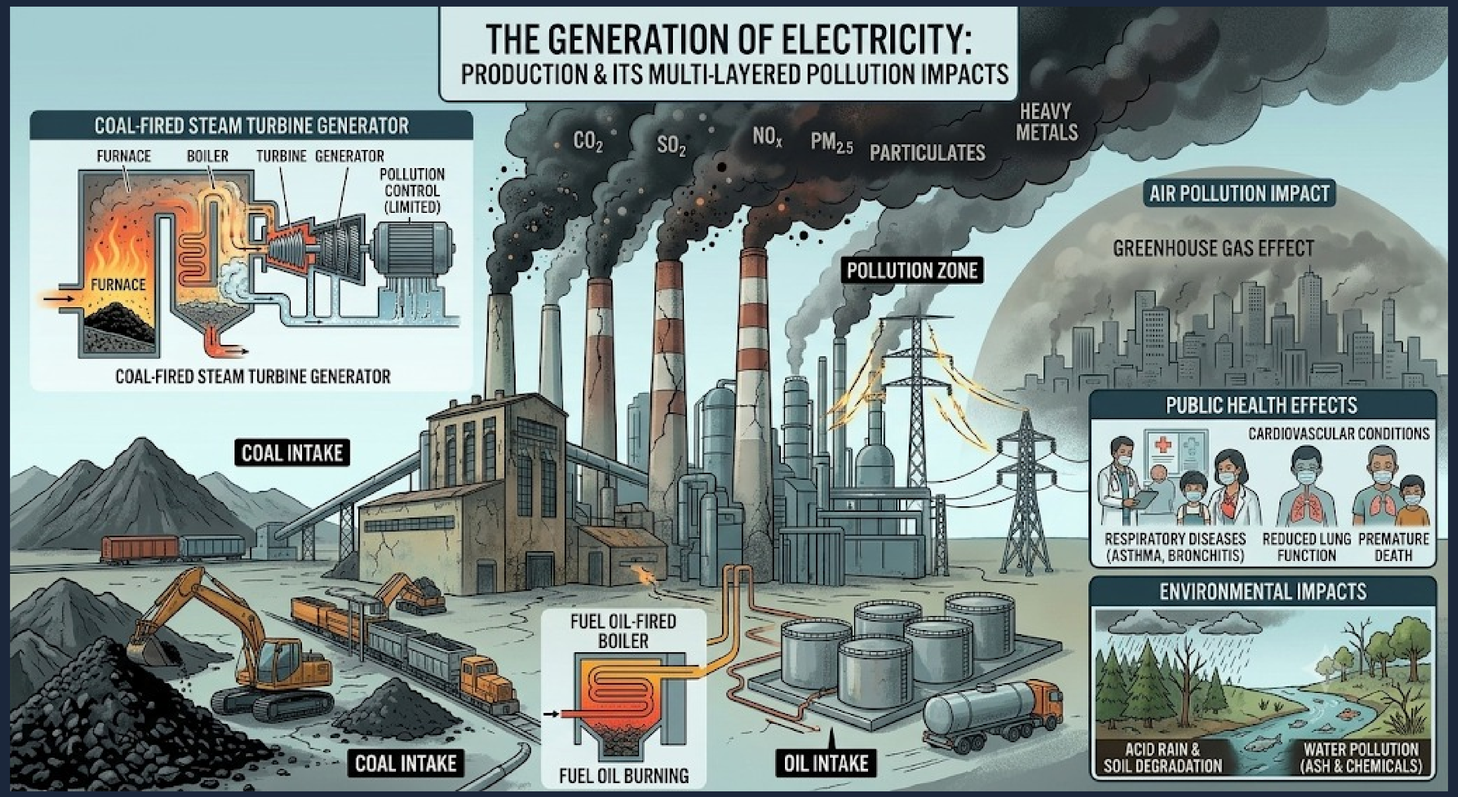
EXISTING SYSTEMS & LIMITATIONS

Current Electricity Sources	Associated Limitations
Coal & Fossil Fuels	High carbon pollution and greenhouse gas emissions
Petroleum & Natural Gas	Limited non-renewable fuel availability and high operating cost
Hydroelectric Power	Requires huge geographical changes and large capital outlay
Solar & Wind Energy	Heavy dependence on weather conditions and high installation cost

FOOTSTEP POWER GENERATION

ENVIRONMENTAL CONTEXT

CONVENTIONAL ELECTRICITY GENERATION & ITS POLLUTION IMPACTS



This visual supports the problem statement and the comparison with conventional electricity sources.

Additional project visual

KEY BENEFITS

ADVANTAGES

- ⭐ Renewable energy source
- ⭐ Eco-friendly solution
- ⭐ Low maintenance
- ⭐ Low operating cost
- ⭐ Easy tile installation
- ⭐ Reduces energy wastage
- ⭐ Portable design
- ⭐ Long operational life
- ⭐ Suitable for smart cities

DEPLOYMENT SCENARIOS

TARGET APPLICATIONS & IMPLEMENTATION

1. Railway Stations
Harvest energy from heavy passenger footfall on platforms and foot overbridges to power station lighting.

2. Shopping Malls
Power indoor ambient lights and digital promotional displays from continuous shopper foot movement.

3. Colleges & Schools
Power hallway emergency lights while demonstrating real-time green energy technology to students.

4. Stadiums
Capture high-density kinetic energy from large crowd movements during sports events and concerts.

5. Dance Floors
Convert rhythmic high-impact foot movement into continuous peak electrical energy for venue lighting.

6. Bus Stands
Energize low-power LED schedule displays and ticketing kiosk indicators from daily waiting commuters.

7. Airports
Power terminal walkway LED direction indicators and sensor arrays from high passenger transit.

8. Metro Stations
Drive automated turnstile indicators and emergency corridor lights in busy underground transit zones.

9. Footpaths & Smart Cities
Provide auxiliary renewable power for smart city street lighting grids and IoT environment sensors.

10. Public Parks
Energize evening pathway safety lamps from dynamic steps generated by daily joggers and visitors.

11. Fitness Centers
Store kinetic energy produced by intense treadmill and aerobic stepping workouts for gym power storage.

FOOTSTEP POWER GENERATION

REAL-WORLD APPLICATION

RAILWAY STATION FOOTSTEP ENERGY HARVESTING



A railway platform is one of the high-footfall locations identified in the Applications slide.

Additional project visual

END OF PRESENTATION

THANK YOU!

Footstep Power Generation Using Arduino

Department of Computer Science